

## MERRITT, BC, Canada

WMO: 715570

Lat: 50.1125N

Lon: 120.7781W

Elev: 591

StdP: 94.42

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -18.6      | -15.2 | -22.0                         | 0.6 | -18.2 | -19.1 | 0.7 | -14.1 | 11.5                  | -3.4 | 10.2 | -3.1 | 1.7                      | 70   | 0.570 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.3                   | 30.3            | 16.5 | 27.9 | 15.9 | 25.8 | 15.8 | 20.2                | 23.4 | 19.1 | 22.0 | 18.2 | 21.2 | 3.2                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 15.0 | 21.6 | 18.1 | 14.0 | 20.6 | 17.1 | 13.1 | 19.5 | 61.0          | 23.3 | 57.0 | 21.4 | 53.5 | 20.8 | 24.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.7           | 8.0  | 6.9 | -21.5                      | 33.5 | 5.1                | 3.1 | -25.2                                              | 35.7 | -28.1      | 37.5 | -31.0      | 39.3 | -34.7      | 41.5 |    |    |
| (5)               |      |     | -22.7                      | 21.5 | 5.0                | 2.0 | -26.3                                              | 22.9 | -29.3      | 24.1 | -32.1      | 25.2 | -35.7      | 26.6 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.4   | -3.9  | -3.1  | 1.6   | 5.9   | 11.0  | 14.8  | 18.5  | 18.7  | 14.9  | 8.7   | 3.2   | -1.8 | (6)  |
| (7)  |                                                                                  | DBStd       | 8.98  | 5.92  | 5.46  | 4.48  | 3.30  | 3.45  | 3.40  | 2.67  | 2.48  | 3.57  | 3.78  | 5.42  | 5.46 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1867  | 432   | 366   | 262   | 129   | 28    | 2     | 0     | 1     | 4     | 71    | 207   | 365  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4071  | 690   | 599   | 520   | 372   | 228   | 117   | 31    | 25    | 112   | 299   | 453   | 623  | (9)  |
| (10) |                                                                                  | CDD10.0     | 928   | 0     | 0     | 1     | 7     | 59    | 145   | 263   | 269   | 151   | 30    | 4     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 91    | 0     | 0     | 0     | 0     | 1     | 9     | 36    | 36    | 9     | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1087  | 0     | 0     | 0     | 5     | 53    | 146   | 431   | 370   | 82    | 1     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 343   | 0     | 0     | 0     | 1     | 11    | 46    | 149   | 118   | 20    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.4   | 2.2   | 2.6   | 2.6   | 3.2   | 2.5   | 2.8   | 2.5   | 2.2   | 2.1   | 2.2   | 2.2   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 353   | 28    | 17    | 21    | 18    | 36    | 46    | 31    | 25    | 26    | 31    | 38    | 36   | (15) |
| (16) |                                                                                  | PrecMax     | 468   | 53    | 33    | 35    | 37    | 78    | 82    | 90    | 60    | 58    | 56    | 74    | 86   | (16) |
| (17) |                                                                                  | PrecMin     | 269   | 13    | 3     | 4     | 5     | 9     | 19    | 1     | 5     | 4     | 12    | 13    | 8    | (17) |
| (18) |                                                                                  | PrecStd     | 53    | 9     | 7     | 7     | 8     | 18    | 17    | 19    | 16    | 16    | 12    | 15    | 17   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 10.1  | 10.1  | 16.0  | 22.7  | 28.1  | 31.2  | 33.7  | 33.2  | 29.4  | 21.1  | 15.3  | 11.6 | (19) |
| (20) |                                                                                  |             | MCWB  | 9.1   | 6.5   | 8.2   | 12.6  | 15.2  | 16.8  | 16.5  | 16.6  | 16.3  | 14.1  | 12.1  | 9.1  | (20) |
| (21) |                                                                                  | 2%          | DB    | 7.9   | 7.9   | 12.6  | 17.0  | 24.4  | 27.5  | 30.4  | 30.0  | 25.6  | 18.5  | 13.7  | 9.2  | (21) |
| (22) |                                                                                  |             | MCWB  | 6.2   | 5.3   | 6.9   | 8.9   | 13.5  | 16.2  | 16.4  | 16.6  | 15.7  | 14.5  | 12.1  | 7.5  | (22) |
| (23) |                                                                                  | 5%          | DB    | 5.8   | 6.2   | 10.3  | 14.1  | 21.1  | 24.1  | 27.9  | 27.2  | 22.9  | 16.6  | 12.1  | 7.7  | (23) |
| (24) |                                                                                  |             | MCWB  | 4.1   | 3.6   | 5.5   | 7.6   | 12.2  | 14.9  | 16.0  | 16.0  | 15.7  | 12.8  | 10.5  | 5.8  | (24) |
| (25) |                                                                                  | 10%         | DB    | 3.8   | 4.4   | 8.2   | 12.0  | 18.1  | 21.5  | 25.3  | 25.0  | 20.8  | 14.8  | 10.5  | 5.9  | (25) |
| (26) |                                                                                  |             | MCWB  | 2.3   | 2.3   | 4.5   | 6.5   | 10.9  | 14.3  | 16.1  | 16.2  | 15.0  | 11.2  | 8.7   | 4.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 9.9   | 7.6   | 9.2   | 12.8  | 16.5  | 20.3  | 21.6  | 21.6  | 19.9  | 18.0  | 13.8  | 9.9  | (27) |
| (28) |                                                                                  |             | MCDWB | 10.2  | 8.5   | 14.3  | 21.7  | 24.7  | 24.5  | 26.0  | 25.1  | 23.1  | 18.8  | 14.4  | 10.5 | (28) |
| (29) |                                                                                  | 2%          | WB    | 6.7   | 5.8   | 7.5   | 9.9   | 14.2  | 18.3  | 20.1  | 20.4  | 18.4  | 15.4  | 12.1  | 8.0  | (29) |
| (30) |                                                                                  |             | MCDWB | 7.5   | 7.7   | 11.5  | 14.9  | 22.8  | 22.5  | 23.4  | 23.1  | 21.1  | 17.1  | 13.0  | 8.7  | (30) |
| (31) |                                                                                  | 5%          | WB    | 4.4   | 4.1   | 6.2   | 8.4   | 12.9  | 16.6  | 18.9  | 19.2  | 17.4  | 13.1  | 10.5  | 5.7  | (31) |
| (32) |                                                                                  |             | MCDWB | 5.9   | 5.8   | 9.7   | 12.9  | 19.7  | 21.5  | 21.6  | 21.8  | 20.1  | 15.4  | 11.4  | 7.0  | (32) |
| (33) |                                                                                  | 10%         | WB    | 2.7   | 2.5   | 5.1   | 7.2   | 11.8  | 15.5  | 17.8  | 18.2  | 16.5  | 11.6  | 8.6   | 3.9  | (33) |
| (34) |                                                                                  |             | MCDWB | 4.1   | 4.5   | 8.2   | 11.1  | 17.5  | 20.7  | 21.3  | 21.3  | 19.4  | 13.9  | 9.9   | 5.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 7.1   | 8.1   | 8.5   | 9.1   | 10.2  | 9.8   | 11.1  | 11.3  | 10.4  | 9.0   | 6.8   | 6.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 8.2   | 9.7   | 12.5  | 14.9  | 17.8  | 16.9  | 19.2  | 18.9  | 15.6  | 11.2  | 7.5   | 8.4  | (36) |
| (37) |                                                                                  |             | MCWBR | 7.2   | 7.4   | 7.4   | 8.1   | 8.1   | 7.3   | 6.8   | 7.5   | 8.0   | 7.4   | 6.2   | 7.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 8.3   | 9.2   | 11.3  | 13.1  | 14.8  | 12.0  | 8.9   | 9.4   | 9.6   | 8.0   | 6.9   | 8.9  | (38) |
| (39) |                                                                                  |             | MCWBR | 7.8   | 7.9   | 7.6   | 7.8   | 7.5   | 6.6   | 5.3   | 5.6   | 5.9   | 6.6   | 6.3   | 8.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.284 | 0.282 | 0.312 | 0.350 | 0.368 | 0.355 | 0.346 | 0.354 | 0.334 | 0.326 | 0.304 | 0.276 | (40) |      |
| (41) |                                                                                  | taud        | 2.375 | 2.438 | 2.428 | 2.350 | 2.356 | 2.418 | 2.473 | 2.457 | 2.492 | 2.491 | 2.436 | 2.410 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 762   | 872   | 900   | 894   | 889   | 902   | 906   | 881   | 865   | 805   | 732   | 721   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 64    | 77    | 93    | 112   | 117   | 111   | 103   | 100   | 87    | 73    | 60    | 53    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.02  | 1.94  | 3.12  | 4.54  | 5.43  | 5.82  | 6.48  | 5.52  | 3.84  | 2.23  | 1.08  | 0.78  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.11  | 0.16  | 0.25  | 0.35  | 0.45  | 0.53  | 0.52  | 0.35  | 0.32  | 0.23  | 0.11  | 0.09  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page